MACHINE LEARNING FOR PREDICTING ABNORMAL CMP RESULTS USING ROUTINE CLINICAL DATA

DANIEL MIDDLETON

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GROUP #3

INTRODUCTION

- Comprehensive Metabolic Panel (CMP):
  - Evaluates kidney, liver, and metabolic function
- Abnormal values lead to critical clinical decisions
- Large-scale clinical data available (MIMIC-IV)
- Opportunity for machine learning

LITERATURE REVIEW

- Ayad et al. (2022)
  - Deep learning models predict ICU lab abnormalities
- Luo et al. (2016)
  - Early work predicting ferritin lab values using ML
- Schober & Vetter (2021)
  - Introduces binary logistic regression in medicine
- MIMIC-IV dataset
  - Large, real-world dataset
  - Widely used for clinical ML research

CURRENT CHALLENGES

- Clinical data is noisy and inconsistent
- Lab “abnormal” flags may be unreliable
- Strong class imbalance
  - Abnormal >> normal
- Complex, nonlinear relationships
- Binary labeling loses severity information

MODEL OBJECTIVE

- Predict abnormal CMP results
- Use:
  - Prior lab values
  - Vitals (BP, BMI)
  - Demographics
  - Diagnoses
- Models:
  - Logistic regression
  - Random forest

FEATURE ENGINEERING

- Prior CMP values
- Delta features (change over time)
- Time between tests
- Diagnosis indicators
- Vitals

CODE DEMONSTRATION

MODEL RESULTS

```
Test AUROC: 0.7584
Test metrics:
sensitivity: 0.8644
specificity: 0.4301
ppv: 0.9244
npv: 0.2823
f1: 0.8934

Classification report:
              precision    recall  f1-score   support

           0     0.2823    0.4301    0.3409     48847
           1     0.9244    0.8644    0.8934    393857

    accuracy                         0.8164    442704
   macro avg     0.6033    0.6472    0.6171    442704
weighted avg     0.8536    0.8164    0.8324    442704
```

Logistic regression

```
Test AUROC: 0.9109
Test metrics:
sensitivity: 0.8387
specificity: 0.8051
ppv: 0.9720
npv: 0.3823
f1: 0.9004

Classification report:
              precision    recall  f1-score   support

           0     0.3823    0.8051    0.5184     48847
           1     0.9720    0.8387    0.9004    393857

    accuracy                         0.8350    442704
   macro avg     0.6771    0.8219    0.7094    442704
weighted avg     0.9069    0.8350    0.8583    442704
```

Random forest

LOGISTIC REGRESSION PREDICTIVE FEATURES

Top positive predictors:

	feature	coefficient	abs_coefficient	odds_ratio
4	num__abnormal_last_5_events	0.542387	0.542387	1.720108
21	num__prev_bun	0.340689	0.340689	1.405916
17	num__prev_glucose	0.236761	0.236761	1.267138
55	cat__admission_type_OBSERVATION ADMIT	0.190596	0.190596	1.209971
35	num__delta_bun	0.185025	0.185025	1.203249
54	cat__admission_type_EW EMER.	0.153970	0.153970	1.166456
46	num__dx_other	0.151692	0.151692	1.163801
0	num__age	0.144000	0.144000	1.154884
36	num__dx_icd10_e11	0.121173	0.121173	1.128820
10	num__prev_alp	0.120669	0.120669	1.128251
50	cat__admission_type_DIRECT EMER.	0.091721	0.091721	1.096059
3	num__prior_abnormal_cmp_count	0.073765	0.073765	1.076554
8	num__prev_alt	0.066729	0.066729	1.069006
13	num__prev_total_bilirubin	0.042914	0.042914	1.043848
40	num__dx_icd10_z79	0.033819	0.033819	1.034397

Top negative predictors:

	feature	coefficient	abs_coefficient	odds_ratio
49	cat__admission_type_AMBULATORY OBSERVATION	-0.490002	0.490002	0.612625
47	cat__sex_F	-0.395037	0.395037	0.673655
48	cat__sex_M	-0.370018	0.370018	0.690722
51	cat__admission_type_DIRECT OBSERVATION	-0.331985	0.331985	0.717498
52	cat__admission_type_ELECTIVE	-0.174848	0.174848	0.839585
14	num__prev_total_calcium	-0.162698	0.162698	0.849848
28	num__delta_total_calcium	-0.157453	0.157453	0.854317
56	cat__admission_type_SURGICAL SAME DAY ADMISSION	-0.137284	0.137284	0.871722
11	num__prev_ast	-0.089570	0.089570	0.914324
18	num__prev_potassium	-0.085017	0.085017	0.918497
34	num__delta_sodium	-0.083941	0.083941	0.919485
15	num__prev_chloride	-0.076379	0.076379	0.926465
39	num__dx_icd10_i25	-0.076188	0.076188	0.926642
30	num__delta_creatinine	-0.061935	0.061935	0.939944
44	num__dx_icd9_401	-0.061391	0.061391	0.940455

RANDOM FOREST PREDICTIVE FEATURES

Top feature importances:

	feature	importance
31	num__delta_glucose	0.179472
4	num__abnormal_last_5_events	0.106798
21	num__prev_bun	0.088715
17	num__prev_glucose	0.081269
35	num__delta_bun	0.067603
16	num__prev_creatinine	0.053682
30	num__delta_creatinine	0.052698
28	num__delta_total_calcium	0.047546
26	num__delta_bicarbonate	0.046584
46	num__dx_other	0.027089
29	num__delta_chloride	0.025623
14	num__prev_total_calcium	0.024738
34	num__delta_sodium	0.022358
2	num__hours_since_last_cmp	0.021498
15	num__prev_chloride	0.014246
25	num__delta_ast	0.014199
24	num__delta_alp	0.013280
32	num__delta_potassium	0.011458
22	num__delta_alt	0.011452
36	num__dx_icd10_e11	0.009073

CONCLUSIONS

- Successfully predicted abnormal CMP results
- Random forest performed best
- Approach supports clinical decision-making
- Future improvements:
  - More specific temporal features
  - Reduce complexity by organizing ICD codes
  - Determine severity levels
  - Explore other ML model types

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